

Environmental Activity & Safety Recommendation Dashboard

Project Proposal

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Target Audience: Louisville, Kentucky residents including individuals planning outdoor activities, fitness enthusiasts, families, commuters, and community members seeking environmental insights and safety guidance.

Technology Stack: Python, PostgreSQL, Dash Plotly, Open-Meteo APIs

Background

This project proposes the development of an interactive Environmental Activity and Safety Recommendation Dashboard that combines weather conditions and air quality indicators to support informed activity planning and environmental safety awareness.

The system will integrate weather and air quality APIs from Open-Meteo and transform raw environmental data into activity recommendations, safety guidance, and interactive visual insights to support better planning decisions.

Business Problem

Many individuals rely on weather applications when planning daily activities; however, environmental quality indicators are often overlooked. Users may unintentionally schedule outdoor activities during periods of poor air quality, extreme weather conditions, or high environmental exposure risks. Traditional weather applications primarily focus on temperature and precipitation forecasts. However, environmental factors such as AQI, particulate matter (PM2.5), UV exposure, humidity, and pollution levels can significantly influence outdoor conditions and user preparedness.

This project addresses the gap by integrating weather forecasts and air quality indicators into a unified environmental recommendation system. The dashboard transforms environmental data into actionable insights, activity recommendations, and safety guidance to support more informed planning decisions.

Objectives

- Build an interactive environmental activity and safety dashboard
- Integrate Open-Meteo Weather and Air Quality APIs
- Develop recommendation algorithms based on weather and environmental conditions
- Generate activity suggestions and environmental safety guidance
- Create alerts for UV exposure, air quality, and weather conditions
- Provide interactive filtering, KPIs, and visual analysis tools

Technical Approach

1. Weather Dashboard

• **Current Conditions:** Display real-time environmental indicators including AQI, PM2.5, PM10, temperature, humidity, wind speed, and UV index to provide users with current environmental conditions.

- **Daily Weather Outlook:** Present forecast summaries to help users understand upcoming environmental and weather patterns for activity planning.

- **Interactive AQI and Weather Charts:** Visualize air quality and weather trends, including AQI, PM levels, UV index, temperature, and precipitation patterns to support environmental awareness and activity planning.

2. Environmental Activity & Safety Recommendation Engine

Air Quality:

- **AQI 0–50:** Favorable conditions for outdoor activities (walking, cycling, hiking)
- **AQI 51–100:** Moderate conditions; limit prolonged outdoor exposure and choose light activities
- **AQI >100:** Reduced outdoor exposure recommended; suggest indoor alternatives (museum visits, shopping, restaurants)

Precipitation:

- **0–30%:** Low precipitation probability; suitable for outdoor activities such as walking and park visits.
- **30–70%:** Moderate probability of rain or light snow; recommend covered outdoor activities and carry weather protection.
- **>70%:** High probability of heavy rain or snow conditions; recommend indoor activities at home such as playing card/board games.

Temperature:

- **50–75°F:** Ideal conditions for outdoor recreation and sports
- **>85°F:** Heat caution with hydration reminders and shaded activities

UV Exposure:

- **UV Index 0–2:** Outdoor activities encouraged
- **UV Index 3–5:** Outdoor activities allowed with SPF 30 sunscreen
- **UV Index 6–10:** Limited outdoor activities with SPF 50 sunscreen
- **UV Index 11+:** Indoor alternatives or outdoor exposure only with SPF 50+ protection

3. Environmental Safety Guidance & Activity Recommendations

- **Humid Conditions:** Wear light clothing and stay hydrated; recommend walking and park visits. Avoid prolonged heat exposure.

- **High Temperatures:** Carry water and wear light clothing; recommend swimming and shaded park activities. Be cautious of heat exhaustion and dehydration.

- **Cold Weather:** Wear warm layered clothing; recommend indoor fitness activities and museum visits. Limit prolonged exposure to cold temperatures.

- **Snow Conditions:** Wear winter protection; be cautious of slippery surfaces and reduced visibility; recommend indoor recreation and winter park walks.

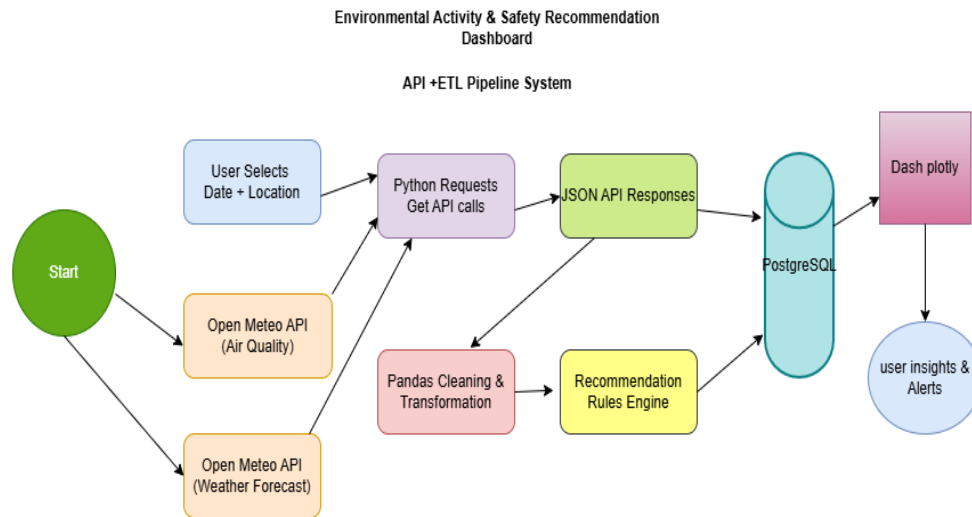
- **Windy Conditions:** Secure loose items and avoid exposed areas; recommend light walking and indoor recreation. Strong winds may affect outdoor comfort.

- **Rain Forecast:** Carry an umbrella; recommend shopping and restaurant visits. Be cautious of wet and slippery conditions.
- **High UV Exposure:** Use SPF 30–50 sunscreen; recommend walking in shaded areas and indoor fitness activities. Avoid prolonged direct sun exposure.
- **Poor Air Quality:** Limit prolonged exposure; recommend museum visits and shopping centers. Reduce intense outdoor activity during elevated pollution levels.

4. Interactive User Interface

- **Dash Plotly Dashboard:** Interactive dashboard for visualizing weather, air quality, and recommendation insights.
- **AQI & Weather Trend Charts:** Display trends for AQI, PM levels, UV index, temperature, and weather conditions.
- **Activity Recommendation Filters:** Filter suggested activities based on environmental and weather conditions.
- **Environmental Condition Cards:** Present key indicators including AQI, temperature, UV index, humidity, and wind conditions.
- **Safety & Recommendation Alerts:** Display activity suggestions, weather precautions, sunscreen guidance, and environmental alerts.

Technical Architecture + Diagram



Backend Components

- **Data Retrieval:** Integrate Open-Meteo Weather and Air Quality APIs to collect environmental and weather data.
- **Data Processing:** Use Pandas for data cleaning, transformation, and formatting of JSON responses.
- **Recommendation Engine:** Apply rule-based logic to generate activity suggestions and environmental safety guidance.
- **Data Storage:** Store processed environmental and weather datasets in a PostgreSQL database for analysis and reporting.

Frontend Components

- **Dashboard Framework:** Dash Plotly for interactive dashboard development and reporting.
- **Visualizations:** AQI trends, PM2.5/PM10 charts, UV exposure trends, weather indicators, environmental KPIs, activity recommendations, and safety alerts.

Data Flow

1. **API Integration:** Retrieve weather and air quality data from Open-Meteo APIs, including temperature, AQI, PM2.5, PM10, UV index, humidity, wind speed, and precipitation.
2. **Data Processing:** Process raw JSON responses using Pandas by cleaning missing values, formatting fields, and organizing the data into structured datasets.
3. **Recommendation Engine:** Apply rule-based logic to generate activity suggestions, safety guidance, and environmental recommendations based on weather and air quality conditions.
4. **Data Storage:** Store cleaned and transformed datasets in PostgreSQL tables for future analysis, reporting, and dashboard use.

5. **Loading Data:** Load processed PostgreSQL data into Dash Plotly to support dashboard visualization and reporting.
6. **User Interface:** Display insights through an interactive Dash Plotly dashboard featuring AQI trends, weather indicators, KPIs, safety alerts, and recommendation cards.
7. **User Interaction:** Allow users to view alerts, filter conditions, and receive activity suggestions and environmental safety guidance based on forecast trends.

Data Sources and APIs

Primary Sources: Open-Meteo Weather API & Air Quality API

- **Endpoint:** <https://api.open-meteo.com/v1/forecast>
- **Location:** Louisville, KY (38.2542°N, 85.7594°W)

Weather Variables Data Parameters

- **Temperature** → temperature_2m — used to recommend suitable activities and weather precautions.
- **Humidity** → relative_humidity_2m — used to evaluate comfort levels and clothing guidance.
- **UV Index** → uv_index — used to generate sunscreen and sun protection recommendations.
- **Wind Speed** → wind_speed_10m — used to assess outdoor activity suitability and safety conditions.
- **Precipitation Probability** → precipitation_probability — used to generate rain alerts and indoor or outdoor activity recommendations.

Air Quality Variables

- **AQI:** Measures overall air quality and supports activity recommendations and environmental safety assessment.
- **PM2.5:** Measures fine particulate pollution to evaluate environmental exposure and outdoor risk conditions.
- **PM10:** Measures larger airborne particles such as dust and pollen that may affect outdoor comfort and activity suitability.
- **Ozone:** Measures ozone concentration levels to support environmental alerts and outdoor exposure recommendations.

Implementation Plan

Phase 1 – Week 1: Finalize Proposal and Research APIs

Complete the project proposal and evaluate Open-Meteo Weather and Air Quality APIs. Review documentation and identify relevant API endpoints, environmental variables, and data parameters.

Phase 2 – Week 2: API Extraction and Ingestion Workflow

Develop Python scripts using Requests to retrieve weather and air quality data. Establish API connections and create the initial data ingestion workflow.

Phase 3 – Week 3: Data Cleaning and CSV Implementation

Use Pandas to clean and transform raw JSON responses into structured datasets. Store processed data in PostgreSQL tables for future retrieval and analysis.

Phase 4 – Week 4: Dashboard Construction and Recommendation Logic

Build an interactive Dash Plotly dashboard and implement activity recommendations, environmental safety guidance, and weather-based alert logic.

Phase 5 – Week 5: Testing and Final Presentation

Conduct system testing to validate data accuracy, dashboard functionality, recommendation outputs, and environmental alerts. Prepare final documentation and presentation materials.

Expected Outcomes

- A fully functional ETL pipeline capable of extracting, transforming, and loading weather and air quality data from Open-Meteo APIs.
- Cleaned and structured environmental datasets stored in PostgreSQL tables for reporting and dashboard integration.
- An interactive Dash Plotly dashboard displaying AQI trends, environmental KPIs, weather indicators, recommendation cards, and safety alerts.
- A rule-based recommendation system that generates activity suggestions and environmental safety guidance using weather and air quality conditions.
- A completed project demonstration showcasing API integration, data transformation, visualization, and environmental recommendation capabilities.

Success Metrics

Technical Metrics

- **API Success Rate:** Maintain a successful API retrieval rate above 99% to ensure reliable weather and environmental data collection.
- **Dashboard Performance:** Dashboard pages should load efficiently and display visualizations without noticeable delays.
- **Data Accuracy:** Ensure retrieved and transformed data accurately reflects current weather and air quality conditions.

User Metrics

- **Recommendation Usefulness:** Measure whether activity suggestions and safety guidance provide meaningful user value.
- **Dashboard Interaction Quality:** Evaluate user interaction with filters, charts, KPIs, alerts, and recommendation features.

Business Metrics

- **Environmental Awareness:** Increase user understanding of how weather and air quality influence daily planning decisions.
- **Decision Support Value:** Provide users with useful activity recommendations and environmental safety guidance.
- **Project Effectiveness:** Demonstrate the practical application of data engineering techniques in solving environmental planning challenges.

Risk Assessment and Mitigation

Technical Risks

- **API Failures:** Data retrieval interruptions may occur due to unavailable API services or connectivity issues.
- **Rate Limiting:** Excessive API requests may exceed usage limits and affect system performance.
- **Missing Values:** Retrieved datasets may contain incomplete or inconsistent environmental information.

Technical Mitigation

- **Logging:** Implement logging to track API requests and monitor workflow performance.
- **Error Handling:** Use exception handling to manage API failures and prevent workflow interruptions.
- **Data Validation:** Validate incoming datasets and handle missing values during preprocessing.

Project Risks

- **Scope Expansion:** Additional dashboard features may increase project complexity and affect completion timelines.
- **Timeline Delays:** Dashboard development and recommendation logic implementation may require additional time.
- **Learning Curve Challenges:** Limited experience with APIs, ETL workflows, Python, or Power BI may slow implementation.

Project Mitigation

- **Feature Prioritization:** Focus on core data pipeline functionality and essential dashboard features.
- **Weekly Milestones:** Follow a structured implementation schedule with achievable goals.
- **Incremental Development:** Build and test one project component at a time to reduce implementation complexity.

Conclusion

This project combines data engineering, environmental analytics, and visualization to create an interactive environmental activity and safety recommendation system. The dashboard transforms weather and air quality data into meaningful insights that support informed activity planning and environmental awareness.

By integrating environmental indicators, activity recommendations, and safety guidance, the system extends beyond traditional weather applications and provides users with actionable insights that support environmental awareness and informed daily planning decisions.

Next Steps

- **Proposal Review:** Review and finalize the project proposal to ensure objectives, architecture, and implementation plans align with project requirements.
- **API Feasibility Assessment:** Evaluate Open-Meteo Weather and Air Quality APIs, verify endpoint availability, and test retrieval of required environmental variables.
- **Prototype Development:** Develop a Minimum Viable Product (MVP) Python script to retrieve weather and air quality data and load test data into PostgreSQL tables.
- **Recommendation Logic Validation:** Test activity recommendation rules and environmental safety guidance thresholds for AQI, UV exposure, temperature, and precipitation conditions.
- **Project Kickoff:** Begin Week 1 development by setting up the project environment, repository structure, Python dependencies, and Dash Plotly workspace configuration.